Vector calculus and differential equation: big picture.

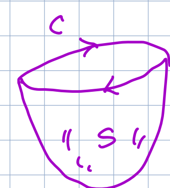
Consider a two-dimensional vector field

$$\vec{u} = \begin{bmatrix} u(x,y,t) \\ v(x,y,t) \end{bmatrix}$$

If a particle moving along a path c in this vector field, then the work done by this vector field is $\int_c \vec{u} \cdot d\vec{r}$

If the vector field is conservative, then the path integral is independent of the choice of the path.

Proof: By stokes theorem



$$\oint_C \vec{u} \cdot d\vec{r} = \iint_S \nabla \times \vec{u} \cdot \hat{n} \, ds$$

Here $\nabla = \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle$ is called gradient operator.

Curl is defined as $\nabla \times F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$

F_x, F_y, F_z are components of the vector $\vec{F}$. (or $\vec{u}$)

For our 2D vector field $\nabla \times \vec{u} = \hat{k} \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right]$

Since $\vec{u}$ is conservative, we have $\nabla \times \vec{u} = 0$

$\boxed{\oint_C u \cdot dr = 0}$ Hence $\int_c u \cdot dr$ is independent of path.

If we drop a particle at position $\vec{x} = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ at $t=0$ in this vector field, the path of the particle is given by

$$\frac{d\vec{x}}{dt} = \vec{u}(x,t)$$

Which is a system of ordinary differential eqution

The vector field itself is governed by a partial differential equation

$$\frac{\partial \vec{u}}{\partial t} = N(\vec{u})$$

where N a partial differential operator.

The divergence of the 2D vector field is

$$\nabla . \vec{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

The divergence of the gradient of a component of $\vec{u}$ is

$$\nabla . \nabla u = \nabla . \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

H.W. Review Stoke's theorem and divergence theorem.

Classification of Partial differential equations

Second order linear PDE

$$\underbrace{Au_{xx} + Bu_{xy} + Cu_{yy}}_{\text{principal part}} + Du_x + Eu_y + Fu = G$$

non-homogeneous

Non linear PDE will have nonlinear terms.

The characteristic equation is

$$\frac{dy}{dx} = \frac{1}{2A}\left(B \pm \sqrt{B^2-4AC}\right)$$

Three cases:

I) $B^2-4AC > 0$, PDE is hyperbolic

Ex: $$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

$A=1, \quad B=0, \quad C=-a^2$

$B^2-4AC = 0^2 - 4(1)(-a^2) = 4a^2 > 0$

There are two characteristic curves

ii) $B^2 - 4AC = 0$, PDE is parabolic

Ex: $\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}$

$A = 0, \; B = 0, \; C = -\nu$

$B^2 - 4AC = 0 - 4(0)\nu = 0$

PDE has a single characteristic curve.

iii) $B^2 - 4AC < 0$, PDE is elliptic.

Ex: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \sin x \cos y$

$A = 1, \; B = 0, \; C = 1$

$B^2 - 4AC = 0 - 4(1)(1) = -4 < 0$

No real characteristic curves

Ex: Classify the PDEs and find characteristic equations

$u_{xx} + 5u_{xy} + 6u_{yy} = 0$

$y^2 u_{xx} - 2y u_{xy} + u_{yy} = u_x + 6y$

A model partial differential equation

$$\boxed{\frac{\partial u}{\partial t} = Lu + q}, \qquad x \in \Omega \subset \mathbb{R}^n, \quad n = 1, 2, 3$$

L is a differential operator

Example:

$$L = -a\frac{d}{\partial x}, \qquad q = 0$$

$$\frac{\partial u}{\partial t} = -a\frac{\partial u}{\partial x}$$

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0 \quad \Rightarrow \text{advection.}$$

Let $u(x,t) = f(x-at)$

$$\frac{\partial u}{\partial t} = -af', \qquad \frac{\partial u}{\partial x} = f'$$

Then

$$-af' + af' = 0$$

Hence $f(x-at)$ is a solution.

At $t=0$, $u(x,0) = f(x)$

This is called initial condition.

At any later time, $u(x,t)$ is $f(x)$ shifted by ct unit.

We can say that a wave propagats at a speed of c

Diffusion

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}, \quad \nu > 0$$

Let $u(x,t) = e^{-\nu t} \sin(x)$

$$\frac{\partial u}{\partial t} = -\nu e^{-\nu t} \sin(x)$$

$$\frac{\partial u}{\partial x} = e^{-\nu t} \cos x$$

$$\frac{\partial^2 u}{\partial x^2} = -e^{-\nu t} \sin x$$

$$\nu \frac{\partial^2 u}{\partial x^2} = -\nu e^{-\nu t} \sin x$$

Hence $u(x,t) = e^{-\nu t} \sin x$ is a solution

Initial Condition is

$$u(x,0) = \sin x$$

Remark: Advection equation is hyperbolic
Diffusion " " parabolic
Laplace equation is elliptic

Aim be able to identify PDEs as hyperbolic, parabolic, elliptic, linear, nonlinear etc.

Solution of differential equation

Let $\frac{du}{dt} = au$, a is a scalar

$u(t) = u_0 e^{at}$, $u(0) = u_0$ is initial condition

Let $\frac{d\vec{u}}{dt} = A\vec{u}$, A is $n \times n$ square matrix

$\vec{u}$ is $n \times 1$ vector

$\vec{u}(t) = \vec{u}_0 e^{At}$

Let $A = \phi \Lambda \phi^{-1}$ be the eigen decomposition

$\vec{u}(t) = \phi e^{\Lambda t} \phi^{-1} u_0$, $\Lambda = diag(\lambda_i), i = 1 \ldots n$

Using the concept of 1st order ODE and eigen decomposition, we solve the system of linear ODEs. Any system of non-linear ODE or PDE can be converted to a linear system before solving it.

Let $\frac{\partial u}{\partial t} = Lu$, L is a linear differential operator.

$u(x,t) = u_0 e^{\lambda t}$, where $Lu = \lambda u$, λ is eigenvector

$u_0(x)$ is to be found

Let λ_k be an eigenvalue

$L\phi_k = \lambda_k \phi_k$

ϕ_k is corresponding eigenfunction.

We see that $u_k = c_k e^{\lambda_k t} \phi_k(x)$ is a solution.

In general, $u(x,t) = \sum_k c_k e^{\lambda_k t} \phi_k(x)$

Take home message:

If a linear PDE is a good approximation of a target PDE, then solution of linear PDE will be an approximate solution of target PDE.

By solving a linear system of ODE, we can get approximate solution of any PDE subject to certain condition.

Conservation law

Consider the density $\rho = \rho(x, t, u, u_x, \cdots)$

and the flux $F = F(x, t, u, u_x, \cdots)$

In general $\vec{u}(\vec{x}, t)$ is a vector field

and flux of a scalar $\rho(\vec{x}, t)$ is

$$\vec{F} = \rho \vec{u}$$

Consider a boundary condition

$$F \to \text{Constant} \quad \text{as } |x| \to \infty$$

Conservation law states that

$$\frac{d}{dt} \int_{-\infty}^{\infty} \rho \, dx = 0$$

In other words $\int_{-\infty}^{\infty} \rho \, dx = \text{Constant}$

This Constant is a conserved quantity and is usually called a constant of motion or an invariant of a corresponding PDE.

Consider the PDE:

$$\frac{\partial \rho}{\partial t} + \frac{\partial F}{\partial x} = 0, \quad F = F(\rho) = \frac{1}{2}\rho^2 \text{ for exampl.}$$

Integrate $\displaystyle\int_{-\infty}^{\infty} \frac{\partial \rho}{\partial t}\, dx + \int_{-\infty}^{\infty} \frac{\partial F}{\partial x}\, dx = 0$

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} \rho\, dx + \lim_{x \to \pm\infty} |F| = 0$$

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} \rho\, dx = 0$$

H.W.

$$\frac{\partial u}{\partial t} - 6u\frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0 \quad \text{(KdV equation)}$$

$u = u(x,t), \quad t \in \mathbb{R}, \quad x \in \mathbb{R}$

i) show that $\displaystyle\int_{-\infty}^{\infty} u\, dx = \text{Constant}$ is an invariant. Find the flux

ii) Multiply the PDE by u and integrali

$$\frac{\partial \frac{1}{2}u^2}{\partial t} - 6u^2\frac{\partial u}{\partial x} + u\frac{\partial^3 u}{\partial x^3} = 0$$

$$\frac{\partial}{\partial t}\frac{1}{2}u^2 + \frac{\partial}{\partial x}\Big(\underbrace{-2u^3 + u\frac{\partial^2 u}{\partial x^2} - \frac{1}{2}\Big(\frac{\partial u}{\partial x}\Big)^2}_{\text{Flux}}\Big) = 0$$

$$\text{flux} = -2u^3 + uu_{xx} - \frac{1}{2}(u_x)^2$$

$\displaystyle\int_{-\infty}^{\infty} \frac{1}{2}u^2\, dx = \text{Constant}$ is an invariant.

Lecture #2, Sep 11, 2025

Fourier analysis method

There are two fundamental approaches for proving stability of numerical methods. One is Fourier analysis (von Neumann analysis).

It applies only to linear constant coefficient PDEs.

The other is energy method, which can be applied to nonlinear PDEs with variable coefficients.

Lax-Richtmyer Equivalence theorem:

Given a well-posed linear initial value problem and its consistent numerical approximation, stability is necessary and sufficient condition for convergence

Proving convergence of any numerical method is often difficult. However stability is easier than showing convergence.

spatial Fourier mode.

Fourier transform is defined as:

$$FT\{f\} = \hat{f}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)\, e^{ikx}\, dx \qquad \text{(forward transform)}$$

e^{ikx} is called Fourier mode

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(k)\, e^{-ikx}\, dk \qquad \text{(Inverse transform)}$$

Here $k = \frac{2\pi}{\lambda}$ is called wavenumber and

λ is called wavelength.

often $\omega = \frac{2\pi}{T}$ is called angular frequence for a period of T if Fourier transform is considered in time domain.

Example: 1)

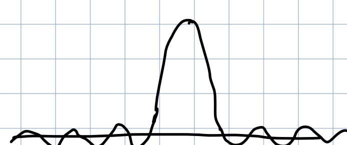
$$f(x) = \begin{cases} 0 & |x| > L \\ 1 & |x| < L \end{cases}$$

$$\hat{f}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)\, e^{ikx}\, dx$$

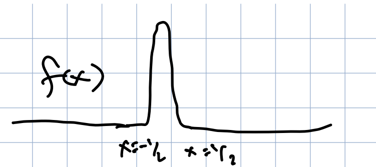
$$\mathrm{sinc}(x) = \frac{\sin x}{}$$

$$= \frac{L}{\pi}\,\frac{\sin Lk}{k}$$

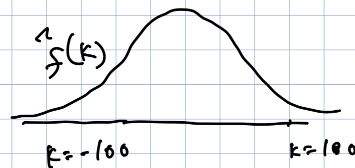
$$= \frac{L}{\pi}\,\mathrm{sinc}(kL)$$



2) $$f(x) = -e^{-\alpha x^2}$$



$$\hat{f}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)\, e^{ikx}\, dx$$



$$= \frac{1}{\sqrt{4\pi\alpha}}\, e^{-\frac{k^2}{4\alpha}}$$

If $f(x)$ has a relatively narrow band in space, it will require a relatively large number of modes (degrees of freedom) for a consistent numerical method.

③ $f(x-\beta)$ where $f(x)$ is any well-defined function and $\beta > 0$

let $$\hat{f}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)\, e^{ikx}\, dx$$

then $$\hat{F}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x-\beta)\, e^{ikx}\, dx$$

$$= e^{ik\beta}\, \hat{f}(k)$$

Prev. lecture $\Rightarrow$

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0$$

$$u(x,0) = f(x)$$

$$u(x,t) = f(x-at)$$

$$= f(x-\beta)$$

H.W. Show that Fourier modes of solution of $\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0$ never decays nor grows for any initial condition, $x \in \mathbb{R}$

If waves travel at constant speed, there is no loss of amplitudes.

4)

Convolution of $f(x)$ and $g(x)$ is defined as

$$(f*g)(x) = \int_{-\infty}^{\infty} f(y)\, g(x-y)\, dy$$

$$= \int_{-\infty}^{\infty} f(x-y)\, g(y)\, dy$$

Fourier transform of convolution becomes multiplication in Fourier domain.

Numerical case: discrete fourier transform (FFTW package)

Let f and g are two sequences defined over an ordered set $\{x_0, x_1, x_2 \cdots\cdots x_N\}$.

Let the convolutional kernel g is defined over a finite interval $[-M, M]$.

For every x_n with $n = 0, 1 \cdots\cdots N$,

$$(f*g)[n] = \sum_{k=-M}^{M} f[n-k]\, g[k]$$

subject to appropriate boundary conditions.

5) ## Fourier transform of derivative

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\partial f}{\partial x} e^{ikx}\, dx$$

$FT\{\frac{\partial f}{\partial x}\} = -ik\hat{f}$

$FT\{\frac{\partial^2 f}{\partial x^2}\} = (-ik)^2\hat{f}$
$= -k^2\hat{f}$

$FT\{\frac{\partial^3 f}{\partial x^3}\} = +ik^3\hat{f}$

$$= \frac{1}{2\pi}\left[f(x)\, e^{ikx}\right]_{-\infty}^{\infty} - \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)(ik)\, e^{ikx}\, dx$$

$$= 0 - \frac{ik}{2\pi}\int_{-\infty}^{\infty} f(x)\, e^{ikx}$$

$$= \boxed{-ik\,\hat{f}(k)}$$

If we have a suitable boundary condition, then Fourier transform of $\frac{\partial f}{\partial x}$ becomes a multiplication in Fourier domain.

Thus
$$\frac{\partial f}{\partial x} = \int_{-\infty}^{\infty} -ik\,\hat{f}(k)\, e^{-ikx}\, dk.$$

Algorithm: Compute derivative.

Let $f = [f_0, f_1, \dots f_N]$

Apply FFT method to compute $\hat{f}$

Multiply by $-ik$ to get $-ik\hat{f}$

Do an inverse FFT to get $\frac{\partial f}{\partial x}$

Summary

1) A finite length convolutional kernel attenuates high frequency modes of a function $f(x)$

2) If $\hat{g}(k) = ik$ is the convolutional kernel in the Fourier domain, the convolution of $g(x)$ and $f(x)$ leads to the derivative of $f(x)$

Ex: 1) $\frac{\partial u}{\partial t} = \frac{\partial^3 u}{\partial x^3}$

Using Fourier transform, show that

$$u(x,t) = \int_{-\infty}^{\infty} \hat{u}(k,0)\, e^{-ik(x+k^2 t)}\, dk$$

2) $$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} + b\frac{\partial^3 u}{\partial x^3} = 0$$

Show that

$$u(x,t) = \int_{-\infty}^{\infty} \hat{u}(k,0)\, e^{-ik(x+(a-bk^2))}\, dk$$

Observe a particular effect of Fourier transform for odd vs even derivatives

Ex: D $\dfrac{\partial u}{\partial t} = \dfrac{\partial^3 u}{\partial x^3}$

Using Fourier transform, show that

$$u(x,t) = \int_{-\infty}^{\infty} \hat{u}(k,0)\, e^{-ik(x-k^2t)}\, dk$$

Sol^n

$$FT_x\left\{\frac{\partial u}{\partial t}\right\} = FT_x\left\{\frac{\partial^3 u}{\partial x^3}\right\},$$

$FT_x\{f(x)\} = \hat{f}(k)$

$$\frac{\partial}{\partial t} FT_x\{u\} = FT_x\left\{\frac{\partial^3 u}{\partial x^3}\right\}$$

$$\frac{\partial \hat{u}(k,t)}{\partial t} = (-ik)^3\, \hat{u}(k,t)$$

$$\boxed{\frac{\partial \hat{u}}{\partial t}(k,t) = ik^3\, \hat{u}(k,t)}$$

$$\hat{u}(k,t) = e^{ik^3t}\, \hat{u}(k,0)$$

$$\frac{du}{dt} = au$$

$$u(t) = u_0 e^{at}$$

$$u(x,t) = \int_{-\infty}^{\infty} e^{ik^3t}\, \hat{u}(k,0)\, e^{-ikx}\, dk$$

$$= \int_{-\infty}^{\infty} \hat{u}(k,0)\, e^{-ik(x-k^2t)}\, dk$$

Remark

1) $\hat{u}(k,t)$ depends on eigenvalue ik^3

2) What happens to a single Fourier mode $\hat{u}(k,t)$

Stability theorem (well-posedness)

Let the Fourier transform of a PDE is

$$\frac{d\hat{u}(k,t)}{\partial t} = A(ik)\,\hat{u}(k,t)$$

Here $A(ik)$ is the Fourier transform of the differential operator.

We have $$\boxed{\hat{u}(k,t) = e^{A(ik)t}\,\hat{u}(k,0)}$$

Since $\|\hat{u}(k,t)\| \leq \|e^{A(ik)t}\|\,\|\hat{u}(k,0)\|$

The Fourier modes will not grow unboundedly if there are constants k and α such that

$$\|e^{A(ik)t}\| \leq k\,e^{\alpha t}$$

Then the eigenvalues of $A(ik)$ defines the region of well-posedness.

H.W. 1) Show that $\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0$, $a>0$

is well-posed.

Hint: $A(ik) = -aik$

2) Using Fourier transform, show that $u(x,t) = f(x-at)$ is the solution of the advection equation.

Dispersion and dissipation

Dispersion refers to the phenomena where waves of different frequencies (or wavenumber) travel at different speeds.

Dissipation refers to the loss of wave amplitude over time, [often due to irreversible energy conversion].

While dispersion and dissipation are important characteristics of the solution of PDE, they are crucial to characterize numerical solution of PDEs. Numerical schemes can artificially cause dissipation and dispersion.

To identify artificial dispersion and dissipation, Fourier analysis and (modified) PDE analysis are crucial.

Dispersion relation:

Let $k \in \mathbb{R}$ and ω is a complex number.

Often e^{ikx} and $e^{i\omega t}$ are called spatial and temporal Fourier modes.

Following the concept from linear ODEs,

$$u(x,t) = e^{i(kx+\omega t)}$$

may be considered a trial solution of a PDE.

Upon substituting into a linear PDE, we must have a relationship between k and ω such as $\boxed{\omega = \omega(k)}$

This is called dispersion relation.

Ex: 1) $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}$, $\quad \omega(k) = k$,

2) $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $\quad \omega(k) = i k^2$ or $i\omega(k) = -k^2$

Hint: Disperssion relation can be obtained for discretized PDE.

1) $\frac{\partial u}{\partial t} = e^{i(kx+\omega t)} \frac{d}{\partial t}(ikx + i\omega t) = e^{i(kx+\omega t)}(i\omega)$

$$\frac{\partial u}{\partial x} = e^{i(kx+\omega t)} \frac{\partial}{\partial x}(ikx+i\omega t) = e^{i(kx+\omega t)}(ik)$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \Rightarrow e^{i(kx+\omega t)} i\omega = e^{i(kx+\omega t)} ik$$

$$\boxed{\omega = k} \Rightarrow \omega = \omega(k) \equiv k$$

For a linear constant coefficient PDE.

The phase speed is

$$c_p(k) = -\frac{\omega(k)}{k}$$

and group speed is

$$c_g(k) = -\frac{\partial \omega}{\partial k}$$

If the phase speed $c_p(k)$ is a function of k, then the solution is dispersive.

Different frequencies travel at diffent speeds

If $\operatorname{Im} \omega > 0$, then

$$|u(x,t)| \leq |\hat{u}_0 e^{i(kx+\omega t)}|$$

$$= |\hat{u}_0 e^{-t \operatorname{Im} \omega(k)}|$$

shows that the solution decays exponentially

The solution is dissipative.

Each Fourier mode decays exponentially

end

Example: Consider the scalar advection equation

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0, \qquad x \in \mathbb{R}, \ t > 0$$

$$u(x,0) = f(x)$$

differential ~~operator~~ operator is $L = c\frac{\partial}{\partial x}$

$$\hat{L}(ik) = \frac{1}{2\pi}\int_{-\infty}^{\infty} c\frac{\partial u}{\partial x} e^{ikx}\, dx$$

$$= -cik\,\hat{u}(k,t)$$

Applying Fourier tranform on to the advection equation, we get

$$\frac{\partial \hat{u}(k,t)}{\partial t} - ick\,\hat{u}(k,t) = 0$$

$$\hat{u}(k,t) = e^{ickt}\,\hat{u}(k,0)$$

where $\hat{u}(k,0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)\, e^{ikx}\, dx$

$$\|\hat{u}(k,t)\| \le \|e^{ickt}\|\,\|\hat{u}(k,0)\|$$

Summary of week 1

- Classification of PDE
- Solution of linear system of ODEs
- Conservation laws for PDEs.
- Dispersion and dissipation by Furier Transform.
- Condition for well-possed PDE

How do you handle even numbers set in FO and Taylor series expansion.